Exp 8a – Determination of the Gas Constant R

INTRODUCTION

Magnesium reacts with acids according to the following net ionic equation.

$$Mg(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g) \quad (1)$$

In this experiment, magnesium will react with hydrochloric acid and the volume of hydrogen gas produced will be measured. The chemical equation for this reaction says that for every 1 mol of Mg used, 1 mol of H_2 gas will form. From this volume, atmospheric pressure, and temperature, the value of "R", the ideal gas constant will be determined.

BACKGROUND

Ideal Gas Law

The ideal gas law (eqn 2) relates the variables of a gas: moles of gas (n), pressure (P), volume (V) and temperature (T). The "R" is the Ideal Gas Constant. Since R is a constant, its value is always the same: 0.082058 L atm/(mol K). In this lab, we will be experimentally determining the Ideal Gas Constant (R) and comparing it against the accepted value.

$$PV = nRT \quad (2)$$

Dalton's Law of Partial Pressures and Vapor Pressures

Dalton's Law of Partial Pressures states that when gases are mixed, the total pressure of the mixture is equal to the sum of the individual partial pressures of the gases.

$$P_T = P_1 + P_2 + P_3 \ldots \quad (3)$$

When a gas is collected "over water" water vapor will be present as part of the mixture because the water will be evaporating to some extent into the gas phase. The pressure of water vapor at different temperatures is known and is recorded on tables in your textbook and in the CRC Handbook of Chemistry and Physics®. The pressure of hydrogen alone can then be determined by applying Dalton's law.

$$P_T = P_{Hydrogen} + P_{Water} \quad (4)$$

Accuracy and Precision

Accuracy is how close your experimental value is to the accepted ("true") value. Precision is how close the experimental values are to each other. Accuracy and precision are largely independent of each other because differences in accuracy or precision are caused by different types of experimental errors.

Systematic errors always occur in the same direction for each repetition and therefore affect accuracy. These may be due to measurement device errors, such as a thermometer that always read 1 °C high, or procedural errors, such as not heating your reaction mixture long enough so that the reaction never reaches completion. The deviation from accuracy (arising from systematic errors(s)) can be quantified by calculating the percent error as shown below.

$$\%\ Error = \frac{experimental\ value - accepted\ value}{accepted\ value} \times 100\%$$

* Other terms are often used in place of the term accepted value, such as theoretical, literature, expected or true value.

ADDITIONAL READING

- Ideal Gas Law, Mixtures of Gases, and Gas Stoichiometry

SAFETY EQUIPMENT REQUIRED FOR LAB

- Goggles
- Lab coat
- Nitrile gloves (provided)

SUPPLY LIST

- 400, 600, or 800-mL beaker
- 50-mL gas collection tube (eudiometer)
- Rubber Stopper
- Steel wool
- Bench Post
- Utility Clamp & Clamp holder
- Fishing line (~15 cm)

CLASSROOM SUPPLY AND REACTANT LIST

- HCl (6 M)
- Magnesium ribbon

PROCEDURE

1. ☐ Obtain a piece of magnesium ribbon 1.4 - 1.8 cm in length and clean it with steel wool.

2. ☐ Measure and record the precise mass of the ribbon. If the mass is more than 0.05 g, trim some of the ribbon off and remeasure the mass.
3. ☐ Tie the Mg Ribbon to one end of a piece of fishing line (~15 cm long).
4. ☐ Set up a bench post, medium clamp, and beaker as shown in Figure 1.

   * *Do NOT include the collection tube in the set-up yet, since it is not yet filled*

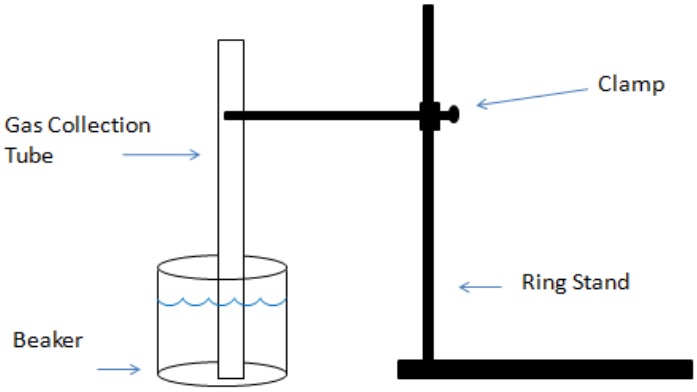
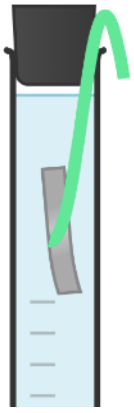


Figure 1: Gas Collection Set-up

5. ☐ Fill a large beaker about halfway with DI water.
6. ☐ Obtain 8 to 10 mL of 6 M hydrochloric acid into your gas collection tube (eudiometer). Do this by placing the eudiometer under the nozzle for bottle-pump dispenser, then slowly doing one complete pump.
7. ☐ Fill the remainder of your tube by *slowly* pouring DI water into your tube all the way to the brim. ** Try to minimize the mixing of water with acid. **
8. ☐ Place the Mg Ribbon into the tube, holding the fishing line over the outer rim. Place the rubber stopper into the tube to hold the fishing line in place.
9. ☐ While holding a finger over the hole in the rubber stopper, invert the tube into your beaker. Remove your finger from the hole once it is under water. Use the clamp as shown in figure 1 to hold the tube upright.



10. ☐ While the magnesium is reacting:
    a. Record the temperature of the water (assumed to be the same as the temperature of the gas).
    b. Record the pressure in the room from the classroom pressure gauge (assumed to be the same pressure as the gas in the tube) *.

    * If the water in the collection tube is not the same as the water level in the beaker, it will cause the pressure in the collection tube to be slightly different than the pressure in the room. For this experiment, we will ignore this difference, which will be a small source of error in the experimental results.

    c. Look up the partial pressure of water in the CRC Handbook®. Record a citation for this book on your data report sheet.

11. ☐ When the magnesium has finished reacting, record the volume of the gas. You will know the reaction is complete when the magnesium ribbon is no longer visible, bubbles stop forming, and the volume of gas stops changing.

CLEAN-UP AND WASTE DISPOSAL

12. ☐ After checking data with your instructor, lift the tube and remove the stopper to allow the solution to drain into the beaker.
13. ☐ With the beaker inside the sink, add sodium bicarbonate (baking soda) to neutralize the acid before dumping it down the drain.
14. ☐ Wash the beaker with soapy water. Rinse with DI water and return to the shelf. There is no need to dry the equipment thoroughly.
15. ☐ Rinse the gas collection tube with soapy water and rinse using your wash bottle (the gas collection tube does not fit in the sink well).
16. ☐ Return equipment.

CALCULATIONS

1. ☐ Use the mass of Mg to calculate the moles of hydrogen gas using the following pathway:

$$g\ Mg \rightarrow mol\ Mg \rightarrow mol\ H_2$$

2. ☐ Use the total atmospheric pressure, and the vapor pressure of water, to calculate the pressure of hydrogen gas according to Dalton's Law of partial pressure.
3. ☐ Convert the pressure of hydrogen gas into atmospheres (atm).
4. ☐ Rearrange the equation $PV = nRT$ to solve for R. Plug in experimental values for P_{H_2}, V, n_{H_2}, and T, ensuring units match the desired units for R: $\frac{L \cdot atm}{mol \cdot K}$
5. ☐ Calculate your percent error

Tip: Do all calculations as if you were in a math class. Only round for significant figures for the final answer to a question. Do NOT use the rounded values in other calculations.



Figure 2:
Hold this black bar behind the eudiometer, below the meniscus to make it easier to read.

DATA AND REPORT – DUE ____ / ____ / ____

SECTION _____

DATE _____

PARTNER _____

* Show your work and keep track of significant figures.

* Record a citation for where you found the vapor pressure of water in the space below.

DATA

Mass of the magnesium ribbon _____

Temperature of water _____

Pressure in the room _____

Vapor pressure of water at your temperature* _____

Volume of gas in collection tube _____

CALCULATIONS

Show your work, complete with units.

1. Moles of hydrogen gas based on mass of Mg used _____

2. Pressure of hydrogen gas (mmHg) _____

3. Pressure of hydrogen gas (atm) _____

4. Experimentally determined R value _____

5. % Error _____

6. What measured value would be directly affected if the magnesium ribbon was not cleaned well?

   This measured value would be (too high, too low, unchanged), which will cause the experimental R value to be (higher, lower, no change).

7. What measured value would be directly affected if the gas collection tube was not fully filled with water and air remained inside the tube as the tube was inverted?

   This measured value would be (too high, too low, unchanged), which will cause the experimental R value to be (higher, lower, no change).

8. Discuss your percent error. Based on your observations, what is at least one source of error specific to your experiment? (Mention the numerical value for your percent error in your response)